Comparative study of data sent over network in English & Sanskrit

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Abstract:

This research focuses on the potential benefits of Sanskrit language and it's strict grammatical structure in field of Computer science and data science. The research's aim is to experiment with Sanskrit data as well as English data (e.g. sentences of same meaning) and analyze the resource usage (e.g. Bytes, Words, Time). The research also provides a graphical analysis of metrics used.

For this purpose, A simple Client – Server model application is used which transmits the given data in both languages and calculates the metrics.

The research shows how English sentences requires extra metadata overhead, containing word order info and on other hand Sanskrit doesn't.

Keywords:

i. **Sanskrit** : An ancient Indian Vedic language
ii. **English** : The language originated in Britain (mid 5-th century AD)
iii. **Metadata** : A "data about data", providing additional info about the other data (e.g. format, source, creation time, etc.
iv. **Bytes** : A unit of digital information in computers, consisting of 8 bits.
v. **Word count** : No. of words a sentence uses.
vi. **Time**
vii. **Data**

1. INTRODUCTION

In the realm of data transmission, efficiency is a critical factor. This research project explores the potential benefits of transmitting data in Sanskrit compared to English, focusing on reducing overhead such as byte size, metadata requirements, and processing time. The project leverages a client-server model to demonstrate how Sanskrit, with its compact and structured linguistic features, can offer advantages over English in data transmission.

1.1 Background

With the increasing demand for efficient data transmission in various fields such as IoT, telecommunications, and cloud computing, optimizing the amount of data transmitted is crucial. Natural languages, being the primary medium of human communication, play a significant role in data representation and transmission. This study investigates how the linguistic properties of Sanskrit can contribute to reducing transmission overhead.

The significant benefits of Sanskrit language over English language in data transfer and management are discussed further.

1.2 Objectives

i. To compare the byte size of sentences in Sanskrit and English.
ii. To analyze the metadata requirements for English and their absence in Sanskrit.
iii. To demonstrate the potential of Sanskrit in reducing data transmission overhead.
iv. To do a comparative study of data sent over network in Sanskrit and English.

1.3 Scope

This research focuses on the transmission of textual data in Sanskrit and English using a client-server model. The study is limited to the analysis of byte size, metadata, and processing time, without delving into other aspects such as encryption or compression

2. **Methodology**

2.1 Client-Server Model

The project implements a **client-server architecture** to simulate data transmission. The client sends sentences in Sanskrit and English to the server, which receives and processes the data.
Java language - Socket programming is used for this purpose.

Client-Side Program :

The client-side program performs the following tasks:

1. Constructs sentences in Sanskrit and English.
2. Measures the byte size of the sentences.
3, Measures the time taken for transmitting the sentences.
4. Sends the data to the server.

Server-Side Program :

The server-side program performs the following tasks:

1. Receives the transmitted data.
2. Reconstructs the sentences.
3. Displays the received data.

2.2 Data Collection

Sanskrit Data: A sentence is broken into individual words and transmitted.

English Data: A sentence is broken into individual words, and metadata (word order) is added for reconstruction.

2.3 Metrics

Byte Size: The total byte size of the transmitted data.

Metadata Overhead: The additional bytes required for metadata in English.

No. of words: The total words required for a sentence in both languages

Processing Time: The time taken to transmit and process the data

2.4 Program Analysis

We would use first an English Strings and then Sanskrit Strings, both forming a sentence of the same meaning.

- ***Sanskrit Sentence:*** सः सूर्यस्य तेजः प्रतापं दृष्ट्वा स्वकान्तं मार्गं लभते
- ***English Sentence***: Seeing the radiance and brilliance of the Sun, He finds his way to his beloved.

i. The Client - Side program will send these strings to Server - side program with distorted order of words.

ii. English sentence will need metadata i.e. an additional string defining the word order. Whereas, Sanskrit sentence will not.

iii. The Client – Side program also counts the time taken by program to run, shown by variable 'TotalTime'. Also, it will count Byte size of Strings using variable 'StrBytes' and 'MDBytes'(additional variable required to count metadata size).

iv. The String array of 'SanskritStrings' and 'EnglishStrings' are array of multiple Strings, each representing a word of sentences.

v. The general (unreserved) port 5000 on local host is used for communication purpose

vi. There are **5 different and complex sentences** used for this project purpose which are listed further.

vii. The sentences provided uses UTF-8 encoding.

3. **Result & Discussions**

3.1 Data Analysis: The data analysis provided below compares the results of operation performed on previously stated various Sanskrit and English statement.

There are 5 different random & complex sentences, Conveying the same meaning in both languages, are used which are listed below:

5 various sentences of English and Sanskrit used are:

A. Seeing the radiance and brilliance of the Sun He finds his way to his beloved (सः सूर्यस्य तेजः प्रतापं दृष्ट्वा स्वकान्तं मार्गं लभते)

B. The large park in the middle of the city offers peace of mind (महानगरस्य मध्यभागे विद्यमानं विशालं उद्यानं मनः शान्ति प्रददाति)

C. The importance of education is essential for the economic development of the nation (राष्ट्रस्य आर्थिकविकासाय शिक्षायाः महत्वं अतीव अनिवार्यमस्ति)

D. The students returned home after hearing the advice of the teachers (विद्यार्थिनः गुरुजनस्य उपदेशं श्रुत्वा स्वगृहं प्रत्यागतः)

E. A sage in the middle of the forest attains great growth in penance (वनस्य मध्ये स्थितः मुनिः तपस्यायां अतिशय वृद्धिं प्राप्नोति)

Program outputs (*Client – side program*):

i. **Initializing Server side program**

```
        String str15 = input.readUTF();$

:!javac Serv_side.java && java Serv_side
Server started, waiting for clients....
```

i. **Running Client side program #Sanskrit sentence**

```
:!javac Cli_Side.java && java Cli_Side
Strings required: 8
Time: 4
String Size in Bytes: 152
```

ii. **Server – side output #Sanskrit sentence**

```
:!javac Serv_side.java && java Serv_side
Server started, waiting for clients....
Client connected
Data: सः सूरयस्य तेजः परतापं दृषट्वा स्वकान्तं मा रगं लभते
```

iii. **Client – side output #English sentence**

```
:!javac Cli_Side.java && java Cli_Side
Strings required: 15
Time: 3
String Size in Bytes: 233
```

iv. **Server – side output #English sentence**

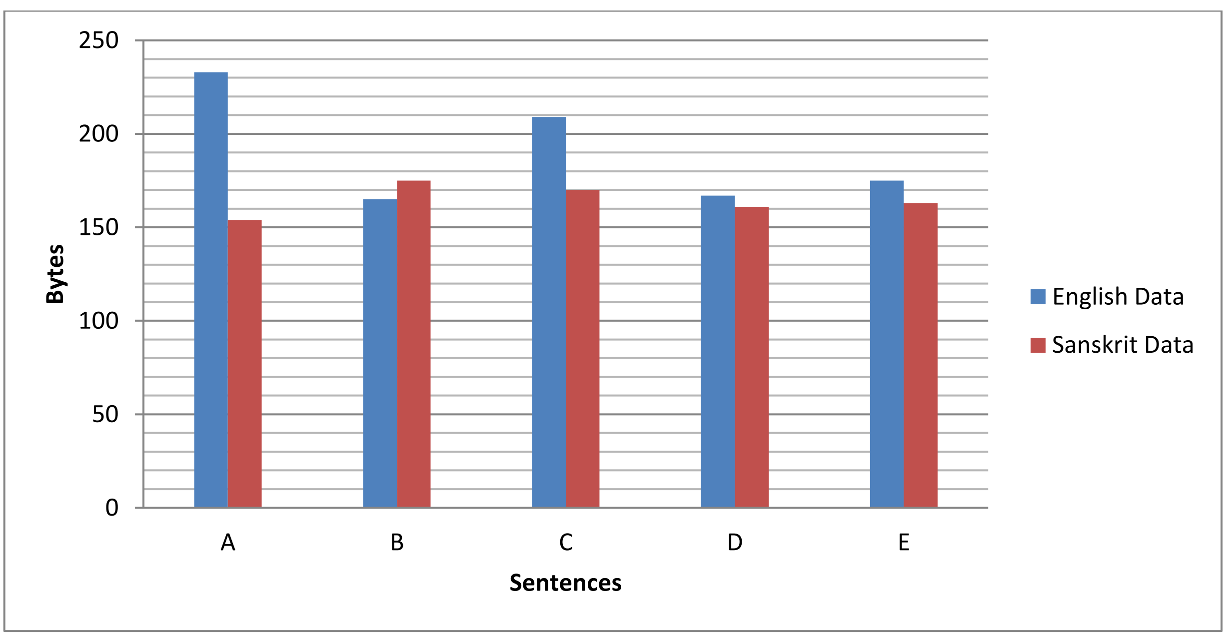
```
:!javac Serv_side.java && java Serv_side
Server started, waiting for clients....
Client connected
Data: Seeing the brilliance and radiance of the Sun He finds his way to his beloved

Metadata: Seeing = 1, The = 2, Brilliance = 3, and = 4, radiance = 5, of = 6, the =
 7, Sun = 8, He = 9, finds =10, his = 11, way = 12, to = 13, his = 14, beloved = 15
```

Outputs for all sentences :

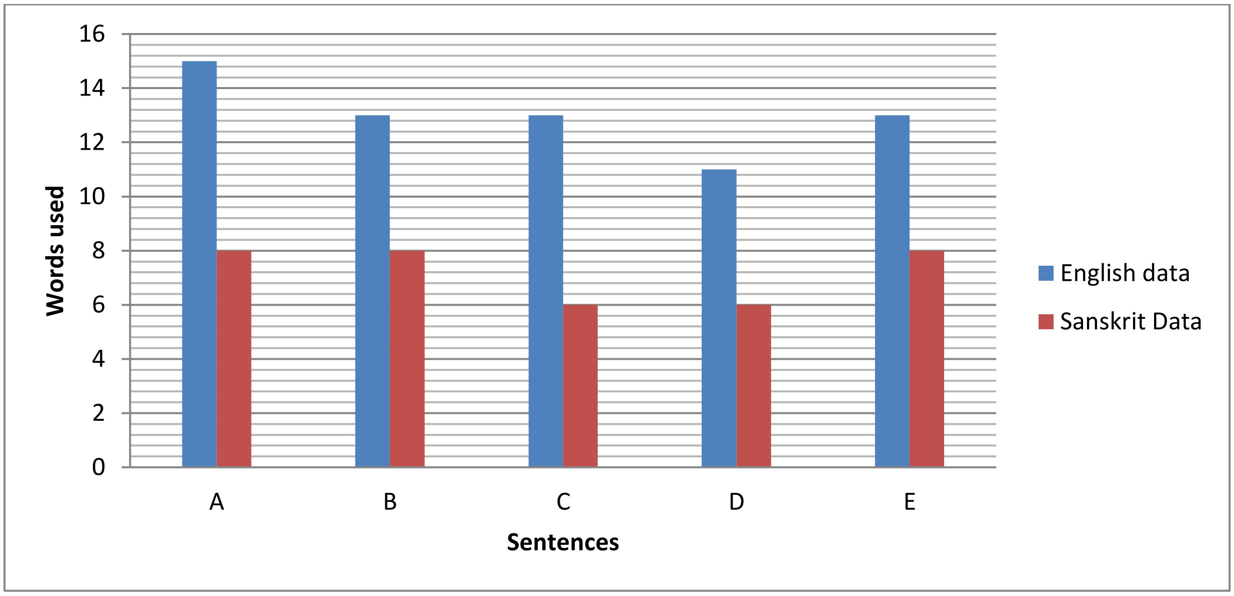
Sentence	English	Sanskrit
i.	Strings required: **15** Time: **3** String size in Bytes: **233**	Strings required: **8** Time: **4** String size in Bytes: **152**
ii.	Strings required: **13** Time: **3** String size in Bytes: **165**	Strings required: **8** Time: **3** String size in Bytes:**175**
iii.	Strings required: **13** Time: **4** String size in Bytes: **209**	Strings required: **6** Time: **3** String size in Bytes: **170**
iv.	Strings required: **11** Time: **3** String size in Bytes: **167**	Strings required: **6** Time: **2** String size in Bytes: **161**
v.	Strings required: **13** Time: **3** String size in Bytes: **175**	Strings required: **8** Time: **3** String size in Bytes: **175**

#Graphical analysis of the given output:

Bytes size analysis:



Words usage:



Time usage:

3.2 Byte Size Comparison Sanskrit vs. English :

The byte size of the Sanskrit sentence was significantly smaller due to its compact word formations. English: The byte size of the English sentence was larger, primarily due to the need for additional metadata to maintain word order.

i. The Byte size of *Sanskrit* revolves around *150 ~ 175* whereas for *English* it's *165 ~ 235*.
ii. The Bytes required in both languages shows a significance difference which is due to overhead of metadata required to send with English data, ultimately increasing the size of data.
iii. Sanskrit Strings can be sent without any care of word order. Hence, even if any malfunctioning or problem occurs, which may affect the data sequence; it won't affect the meaning of the data.
iv. If any problem to the packet itself occurs, due to the strict grammar rules, we can guess the fault where it is occurred.

3.3 Metadata Overhead Sanskrit vs. English :

Sanskrit: No metadata was required, as the sentence structure is inherently clear.

English: Metadata was necessary to ensure the correct reconstruction of the sentence, adding to the overhead.

3.4 Word count :

English: Word count ranges between *11 ~ 15* words

Sanskrit: Word count ranges between *6 ~ 8* words

Hence, Data in Sanskrit can save significant amount of storage and processing power as well as increasing speed and overall quality of data transmission.

3.5 Processing Time

Although processing time was approx. same (*3 ~ 4 ms*) for given sentences on same machine, But, The processing time for Sanskrit was marginally lower, as the server did not need to interpret metadata.

The significant difference of this may be seen in relatively bigger dataset.

Note: Processing time may vary on different machines

3.6 Discussion

The results indicate that transmitting data in Sanskrit can reduce overhead by minimizing byte size and eliminating the need for metadata. This efficiency can be particularly beneficial in resource-constrained environments.

When the relatively large datasets needs to be transmitted or to be compressed, Sanskrit can be an efficient option because:

- Sanskrit focuses on word's grammar rather than order, Useful for eliminating metadata overhead in data transfer
- Sanskrit doesn't requires various articles of speech unlike English (e.g., 'the', 'a', 'an', etc.) or separate words to show relation like 'of', 'for', 'to', etc. Sanskrit encodes the meaning in much lesser words, due to it's strict grammatical structure, Hence it's an efficient way in fields like data compression.
- Few researches also suggests that Sanskrit can be a *revolutionary programming language* due to it's grammar.

4. **Conclusion**

- This research demonstrates that Sanskrit, with its compact and structured linguistic features, offers significant advantages over English in data transmission. By reducing byte size and eliminating metadata requirements, Sanskrit can help optimize network performance and reduce overhead. Future work could explore the application of this approach in real-world scenarios, such as IoT devices or low-bandwidth communication systems.
- It’s an efficient way to manage the data and perform operations on it when it comes to resource – constrained environments.
- It’s cost effective, as it saves a lot of space due to its compact structure. It’s resource effective and can save lots of bandwidth when it comes to networking.
- It can open new ways in programming world if used as a programming language.

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5. **References**

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